ORGANIC CHEMISTRY · ASYNCHRONOUS ACTIVITY 02

One Ninja, Four Views

One Molecule, Four Representations:
From Molecular Formula to Bond-Line Structure

James Joseph A. Pulvinar
BSChE-2A



Froakie says hi

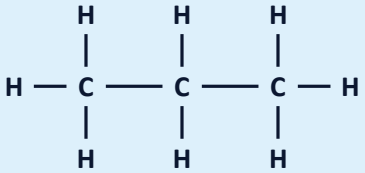
CARD 01 — PROPANE

Molecular Formula: C_3H_8

1 MOLECULAR FORMULA

$$C_3H_8$$

2 LEWIS STRUCTURE



3 CONDENSED STRUCTURAL FORMULA

$$CH_3CH_2CH_3$$

4 BOND-LINE FORMULA



✓ Atom Count Verified

✓ Valence Checked

✓ Bond order checked

✓ Same connectivity confirmed

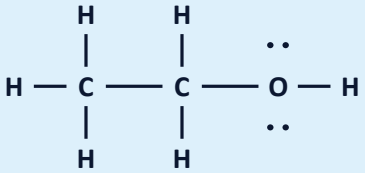
CARD 02 — ETHANOL

Molecular Formula: C_2H_6O

1 MOLECULAR FORMULA

$$C_2H_6O$$

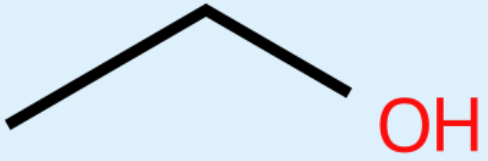
2 LEWIS STRUCTURE



3 CONDENSED STRUCTURAL FORMULA

$$CH_3CH_2OH$$

4 BOND-LINE FORMULA



✓ Atom Count Verified

✓ Valence Checked

✓ Bond order checked

✓ Same connectivity confirmed

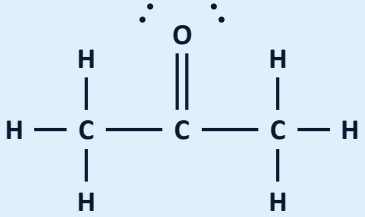
CARD 03 — ACETONE (PROPAN-2-ONE)

Molecular Formula: C_3H_6O

1 MOLECULAR FORMULA

$$C_3H_6O$$

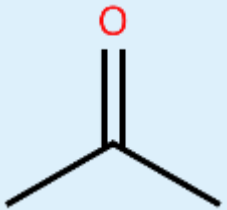
2 LEWIS STRUCTURE



3 CONDENSED STRUCTURAL FORMULA

$$CH_3COCH_3$$

4 BOND-LINE FORMULA



✓ Atom Count Verified

✓ Valence Checked

✓ Bond order checked

✓ Same connectivity confirmed

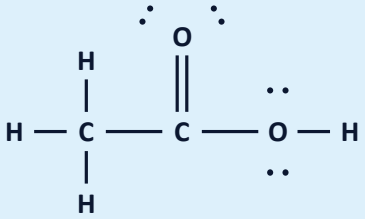
CARD 04 — ACETIC ACID (ETHANOIC ACID)

Molecular Formula: $C_2H_4O_2$

1 MOLECULAR FORMULA

$$C_2H_4O_2$$

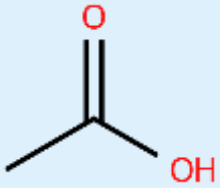
2 LEWIS STRUCTURE



3 CONDENSED STRUCTURAL FORMULA

$$CH_3COOH$$

4 BOND-LINE FORMULA



✓ Atom Count Verified

✓ Valence Checked

✓ Bond order checked

✓ Same connectivity confirmed

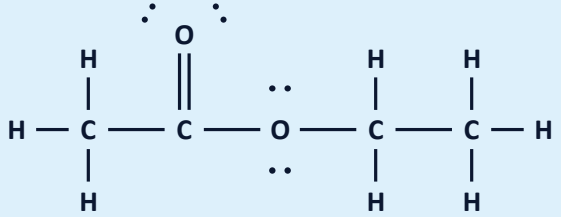
CARD 05 — ETHYL ACETATE (ETHYL ETHANOATE)

Molecular Formula: $C_4H_8O_2$

1 MOLECULAR FORMULA

$$C_4H_8O_2$$

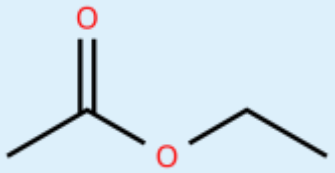
2 LEWIS STRUCTURE



3 CONDENSED STRUCTURAL FORMULA

$$CH_3COOCH_2CH_3$$

4 BOND-LINE FORMULA



✓ Atom Count Verified

✓ Valence Checked

✓ Bond order checked

✓ Same connectivity confirmed

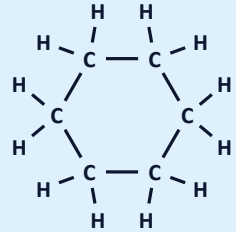
CARD 06 — CYCLOHEXANE

Molecular Formula: C_6H_{12}

1 **MOLECULAR FORMULA**

$$C_6H_{12}$$

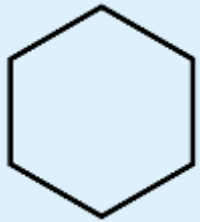
2 **LEWIS STRUCTURE**



3 **CONDENSED STRUCTURAL FORMULA**

$$(CH_2)_6$$

4 **BOND-LINE FORMULA**



✓ Atom Count Verified

✓ Valence Checked

✓ Bond order checked

✓ Same connectivity confirmed

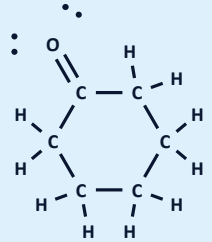
CARD 07 — CYCLOHEXANONE

Molecular Formula: $C_6H_{10}O$

1 **MOLECULAR FORMULA**

$$C_6H_{10}O$$

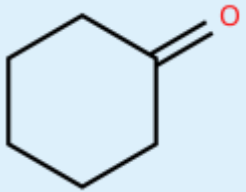
2 **LEWIS STRUCTURE**



3 **CONDENSED STRUCTURAL FORMULA**

$$(CH_2)_5CO$$

4 **BOND-LINE FORMULA**



✓ Atom Count Verified

✓ Valence Checked

✓ Bond order checked

✓ Same connectivity confirmed

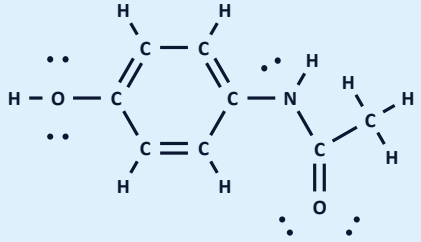
CARD 08 — ACETAMINOPHEN (PARACETAMOL)

Molecular Formula: $C_8H_9NO_2$

1 MOLECULAR FORMULA

$$C_8H_9NO_2$$

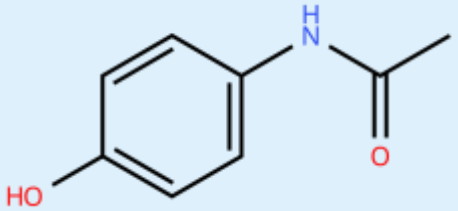
2 LEWIS STRUCTURE



3 CONDENSED STRUCTURAL FORMULA

$$HOC_6H_4NHCOCH_3$$

4 BOND-LINE FORMULA



✓ Atom Count Verified

✓ Valence Checked

✓ Bond order checked

✓ Same connectivity confirmed

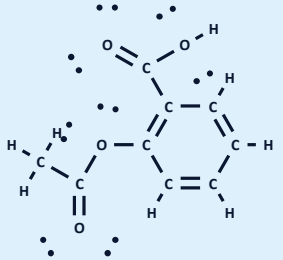
CARD 09 — ASPIRIN (ACETYLSALICYLIC ACID)

Molecular Formula: $C_9H_8O_4$

1 **MOLECULAR FORMULA**

$$C_9H_8O_4$$

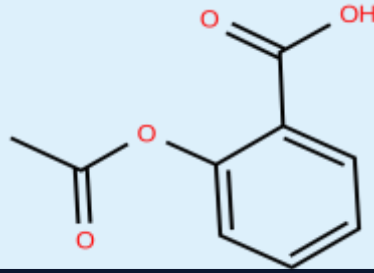
2 **LEWIS STRUCTURE**



3 **CONDENSED STRUCTURAL FORMULA**

$$CH_3COOC_6H_4COOH$$

4 **BOND-LINE FORMULA**



✓ Atom Count Verified ✓ Valence Checked ✓ Bond order checked ✓ Same connectivity confirmed

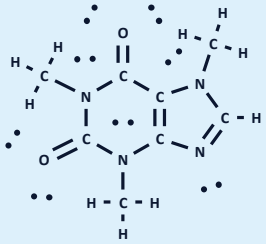
CARD 10 — CAFFEINE

Molecular Formula: $C_8H_{10}N_4O_2$

1 **MOLECULAR FORMULA**

$$C_8H_{10}N_4O_2$$

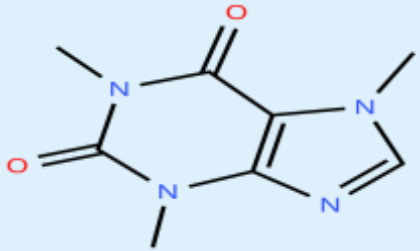
2 **LEWIS STRUCTURE**



3 **CONDENSED STRUCTURAL FORMULA**

$C_8H_{10}N_4O_2$ or
$C_5HN_4O_2(CH_3)_3$

4 **BOND-LINE FORMULA**



✓ Atom Count Verified ✓ Valence Checked ✓ Bond order checked ✓ Same connectivity confirmed